

TOPS Technologies

N+ Network Security - Assessment

Theme: Startup Office IT Setup | Total Time: 3 Hours

SECTION A: Scenario-Based Concept Application

Q1. CPU, RAM, and Storage Selection for a Startup

CPU Selection:

For concurrent development and video conferencing, choose a multi-core processor. Intel Core i7/i9 or AMD Ryzen 7/9 are ideal. Higher clock speed ensures smooth multitasking. Always verify socket compatibility with the motherboard before purchasing.

RAM Selection:

- Minimum 16GB DDR4 RAM; 32GB recommended for heavy workloads.
- Match RAM speed with motherboard's supported frequency.
- Use dual-channel configuration for better bandwidth.
- Ensure RAM type (DDR4/DDR5) matches the motherboard slot specification.

Storage Selection:

- Use an NVMe SSD (512GB-1TB) for OS and applications - far faster than SATA SSDs or HDDs.
- Add an HDD (1TB+) for bulk project file storage.
- Verify interface compatibility: NVMe uses M.2 slot; SATA SSD uses SATA port.

Bottleneck Prevention:

Balance all three components. A fast CPU paired with slow RAM or an HDD will create bottlenecks. Use PCIe Gen 4 NVMe drives if the motherboard supports it. Verify TDP of the CPU against the cooling solution.

Q2. Resource Allocation and Hardware Bottlenecks from Antivirus Software

When antivirus software runs in the background, it continuously performs real-time file scanning, heuristic analysis, and scheduled deep scans, consuming significant CPU cycles and disk I/O.

How Bottlenecks Occur:

- CPU: Antivirus uses additional threads, reducing availability for user applications.
- RAM: Background processes occupy RAM, forcing the OS to use pagefile (virtual memory on disk).
- Disk I/O: Constant scanning causes read/write congestion, slowing file access.
- Network: Antivirus cloud-lookup features consume bandwidth.

Performance Degradation Chain: When RAM is full, the OS uses paging/swapping to disk. The disk becomes a bottleneck, the CPU waits for data, and overall system slows drastically. This is called thrashing. On systems with HDDs instead of SSDs, this degradation is far more severe.

Solution: Allocate dedicated resources, schedule scans during off-hours, and ensure sufficient RAM so paging is minimized.

Q3. Systematic Troubleshooting for Peripheral Hardware Failures

Step 1 - Identify the Problem: Document exactly when and where the device fails (which workstation, which OS version). Check if the device works on one workstation but not another to isolate a software/driver vs. hardware issue.

Step 2 - Check Device Manager / dmesg: Look for yellow warning icons or error codes. Note the exact error code for reference.

Step 3 - Test the Hardware: Connect the peripheral to a known-working system. Try a different USB port or cable to rule out physical issues.

Step 4 - Driver Investigation: Uninstall the current driver completely. Download the latest driver from the manufacturer's official website. If the latest driver fails, roll back to a previously working version.

Step 5 - OS Compatibility Check: Confirm the driver version supports the installed OS. Check Windows Update as OS patches can sometimes break peripheral compatibility.

Step 6 - Check for OS Conflicts: Disable conflicting services or recently installed software. Run the device on a clean boot (msconfig) to isolate conflicts.

Step 7 - Test on Another OS: Boot a Linux Live USB and test the peripheral. If it works, the issue is Windows-specific .

Step 8 - Document and Resolve: Apply the fix uniformly across all affected workstations. Update IT documentation with the resolution steps.

Q4. Role of UPS and Surge Protection in Data Integrity

Uninterruptible Power Supply (UPS):

A UPS provides battery backup power that instantly activates when mains power fails or fluctuates. It serves three critical functions:

- Graceful Shutdown Time: Gives the OS enough time to save files and shut down properly, preventing mid-write corruption.
- Voltage Regulation: Built-in AVR (Automatic Voltage Regulation) smooths out voltage sags and swells.
- File System Integrity: Prevents interrupted disk writes that cause journal corruption on NTFS or ext4 file systems, protecting the MBR and partition table.

Surge Protectors:

Surge protectors use Metal Oxide Varistors to absorb voltage spikes from lightning or grid switching. Without surge protection, spikes can physically destroy the PSU, motherboard, or HDDs, and corrupt data in NVMe/SSDs.

Combined Role: UPS handles outages and brownouts. Surge protectors handle spikes and transients (over-voltage). Together, they ensure the disk is never left mid-write, file systems remain consistent, and hardware is protected.

SECTION B: Practical Application

Task 1 - Workstation Assembly Sequence

1. Prepare the case: remove side panel, install standoffs.
2. Install CPU onto motherboard (align arrow markers, no force).
3. Apply thermal paste; attach CPU cooler.
4. Insert RAM sticks (use slots A2/B2 for dual-channel).
5. Install M.2 NVMe SSD onto motherboard slot.
6. Mount motherboard into case and secure with screws.
7. Install PSU into case and route cables.
8. Install GPU into PCIe x16 slot.
9. Connect SATA drives (HDD/SSD) and data cables.
10. Connect all power connectors (24-pin ATX, 8-pin CPU, PCIe power).
11. Connect front-panel connectors (power button, USB, audio headers).
12. Cable management and case closure.
13. Power on and verify POST.

Task 2 - Troubleshooting RAM Mismatch in BIOS/UEFI

Symptoms: System beeps on boot, RAM not fully detected, or BIOS shows wrong speed.

- Enter BIOS/UEFI (press DEL or F2 on startup).
- Navigate to Advanced Memory Settings or DRAM Configuration.
- Check detected RAM speed vs. rated speed (e.g., installed 3200MHz but BIOS shows 2133MHz).
- If mismatch persists, manually set voltage, timings, and frequency.
- Verify both sticks are seated in correct slots.
- Save and Exit (F10) - confirm system POSTs successfully and detects full RAM.

Task 3 - LAN Topology in Cisco Packet Tracer (Class C Subnet)

Network: 5 PCs + 1 Printer on subnet 192.168.1.0/24

IP Address Scheme:

- PC1: 192.168.1.10 | PC2: 192.168.1.11 | PC3: 192.168.1.12
- PC4: 192.168.1.13 | PC5: 192.168.1.14 | Printer: 192.168.1.20
- Subnet Mask: 255.255.255.0 | Default Gateway: 192.168.1.1 (Switch)

Steps in Packet Tracer:

- Add 1 Switch (2960), 5 PCs, and 1 Printer.
- Connect all devices to the switch using straight-through copper cables.
- Assign static IPs to each device as listed above.
- Verify connectivity using ping 192.168.1.x from each PC.
- Success Criteria: All pings succeed.

Task 4 - Clean OS Installation with Manual Disk Partitioning

- Boot from Windows/Linux ISO (set boot priority in BIOS to USB).
- Select Custom Installation (not Upgrade).
- Format and install OS on the root partition.
- Complete setup: region, user account, network.
- Startup Optimization - Windows: Task Manager > Startup > Disable unnecessary items. Linux: systemctl disable for unused services. Enable Fast Boot in BIOS if supported.
- Verify OS boots without errors.

SECTION C: Integrated Case Study - Nexus Innovations

Scenario: Three rendering workstations pass morning POST checks but restart randomly during heavy afternoon workloads. The units are stored in poorly ventilated desk cabinets.

Q1. Three Hardware Components Most Likely Causing Random Reboots Under Load

- CPU: Overheating due to sustained high utilization during rendering; thermal throttling leads to protective shutdown.
- GPU: Rendering workloads heavily stress the GPU; insufficient cooling triggers a system-protective shutdown.
- PSU (Power Supply Unit): Under heavy load, an undersized or aging PSU cannot deliver stable wattage; voltage drops cause reboots. Capacitors degrade faster under heat.

Q2. Why Systems Pass POST in the Morning but Fail in the Afternoon

In the morning, systems and their environment are at ambient (cool) temperature. Components operate within safe thermal limits, so POST completes successfully.

By afternoon, after hours of rendering in poorly ventilated desk cabinets, heat accumulates inside the enclosure. CPU and GPU temperatures rise progressively. When temperatures exceed safe thresholds, the BIOS thermal protection mechanism triggers an automatic shutdown/reboot to prevent permanent hardware damage.

Root Cause: Thermal saturation builds over time, and the enclosed cabinets trap heat with no airflow to dissipate it.

Q3. Using Stress-Test Tools to Confirm Thermal or Power Issues

Tools to use:

- CPU stress: Prime95 or AIDA64
- GPU stress: FurMark or 3DMark
- Combined system stress and monitoring: HWiNFO64 or OCCT

Process:

- Install HWiNFO64 to monitor real-time CPU temp, GPU temp, voltages, and fan RPM.
- Run Prime95 (Blend test) to max out CPU and RAM - watch for temperatures climbing above 90 degrees C.

- Run FurMark to stress the GPU - monitor GPU temp; if it exceeds 85 degrees C and the system reboots, a thermal issue on the GPU is confirmed.
- Monitor 12V, 5V, and 3.3V rails in HWiNFO64 during stress - if voltages drop below 5% of rated value, the PSU is failing under load.
- Compare morning vs. afternoon readings - significantly higher afternoon temps confirm thermal accumulation as the root cause.

Q4. Three Specific Remediation Steps

- Step 1 - Improve Airflow and Ventilation: Remove workstations from enclosed desk cabinets and place them in open, ventilated spaces. If cabinets must be used, cut ventilation holes or install cabinet fans to create active airflow intake at bottom, exhaust at top.
- Step 2 - Clean and Upgrade Internal Cooling: Open each workstation and clean dust from all heatsinks, fans, and vents using compressed air. Replace dried thermal paste on the CPU and GPU with fresh high-quality paste. Upgrade CPU coolers to higher-capacity tower coolers if the stock cooler is insufficient for sustained rendering workloads.
- Step 3 - Configure Thermal Monitoring and BIOS Alerts: In BIOS/UEFI, set fan curves to ramp up cooling at lower temperature thresholds. Deploy HWiNFO64 or SpeedFan on each workstation to log temperatures continuously, enabling early detection before thermal shutdowns occur.

Q5. Technical Justification for PSU Upgrade

A PSU upgrade may be necessary if:

- Voltage monitoring during stress testing shows 12V or 5V rails dropping outside the 5% tolerance, indicating the PSU cannot sustain peak wattage.
- Workstations were recently upgraded with higher TDP CPUs/GPUs without a corresponding PSU upgrade, leaving insufficient headroom.
- The PSU is older (3+ years) and capacitors have degraded, reducing efficiency and stable output under heat.

A PSU upgrade may not be necessary if:

- Voltage rails remain stable during stress testing and the root cause is purely thermal (heat buildup from poor ventilation).
- The PSU wattage already provides 20-30% headroom above the system's peak consumption.
- Fixing the cooling and airflow completely resolves the random reboots, confirming the PSU was never the limiting factor.

Conclusion: First remediate the thermal environment. Only replace the PSU if voltage instability is confirmed through monitoring, or if the wattage headroom is genuinely insufficient for the workload.